Olist Brazilian E-Commerce Sales Data Analysis Using SQL Server and Python

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1. Introduction

E-commerce has become one of the fastest-growing sectors worldwide, generating massive amounts of transactional data every day. Analyzing this data allows businesses to better understand customer behavior, optimize operations, improve customer satisfaction, and increase revenue.

This project analyzes the Brazilian E-Commerce Public Dataset provided by Olist. The dataset contains approximately 100,000 orders placed between 2016 and 2018 across multiple marketplaces in Brazil. Microsoft SQL Server was used for data storage and querying, while Python was used for data visualization and analysis.

The primary objective of this project is to discover meaningful business insights through SQL queries involving JOIN and GROUP BY operations and present those insights through visualizations.

2. Dataset Description

The dataset consists of nine CSV files containing information about customers, orders, products, sellers, payments, reviews, and geographical data.

Dataset Files

- olist_customers_dataset.csv
- olist_geolocation_dataset.csv
- olist_order_items_dataset.csv
- olist_order_payments_dataset.csv
- olist_order_reviews_dataset.csv

- olist_orders_dataset.csv
- olist_products_dataset.csv
- olist_sellers_dataset.csv
- product_category_name_translation.csv

Main Features

- Customer information
- Order details
- Product information
- Seller information
- Payment methods
- Customer reviews
- Geographic locations
- Product category translations

The dataset provides a complete view of the e-commerce process, from purchase to delivery and customer feedback.

3. Database Design

Microsoft SQL Server was used to create and manage the database.

The following tables were created:

- customers
- orders
- order_items
- order_payments
- order_reviews
- products
- sellers
- geolocation
- category_translation

Relationships between tables were established through common identifiers such as:

- customer_id
- order_id

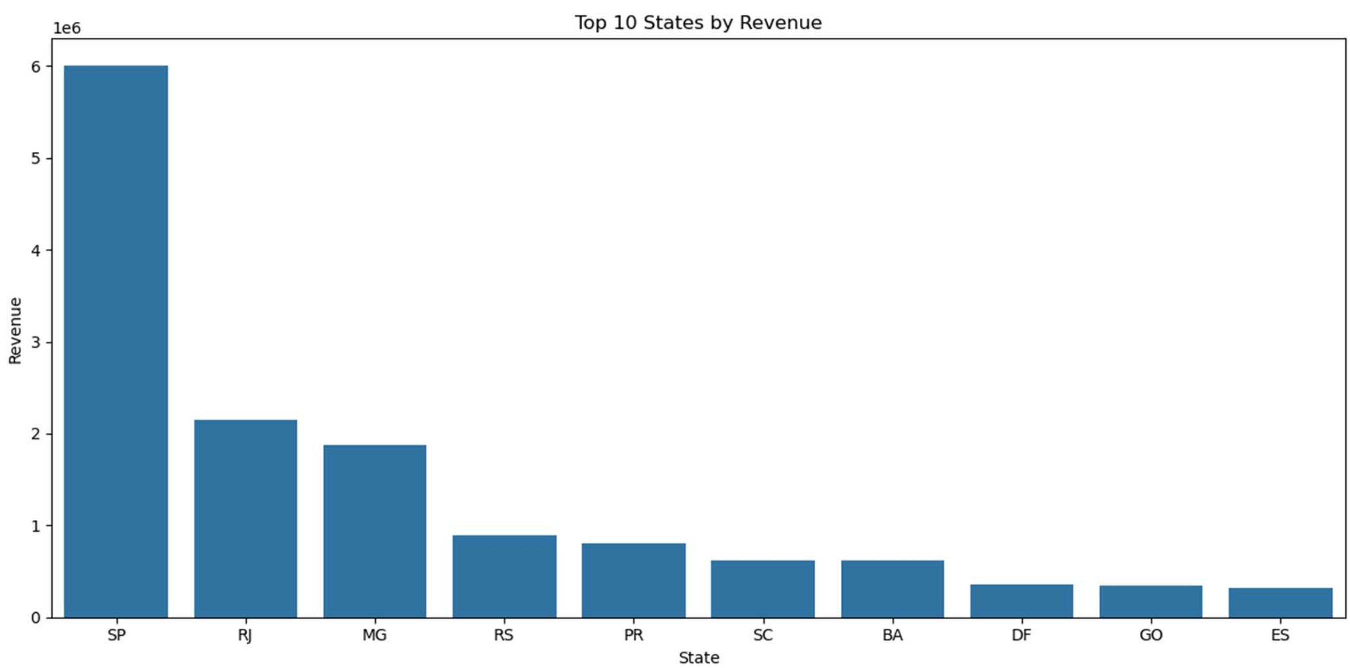
- product_id
- seller_id

These relationships enabled efficient analysis using SQL JOIN operations.

5. Visualizations

Python libraries including Pandas, Matplotlib, and Seaborn were used to create visualizations.

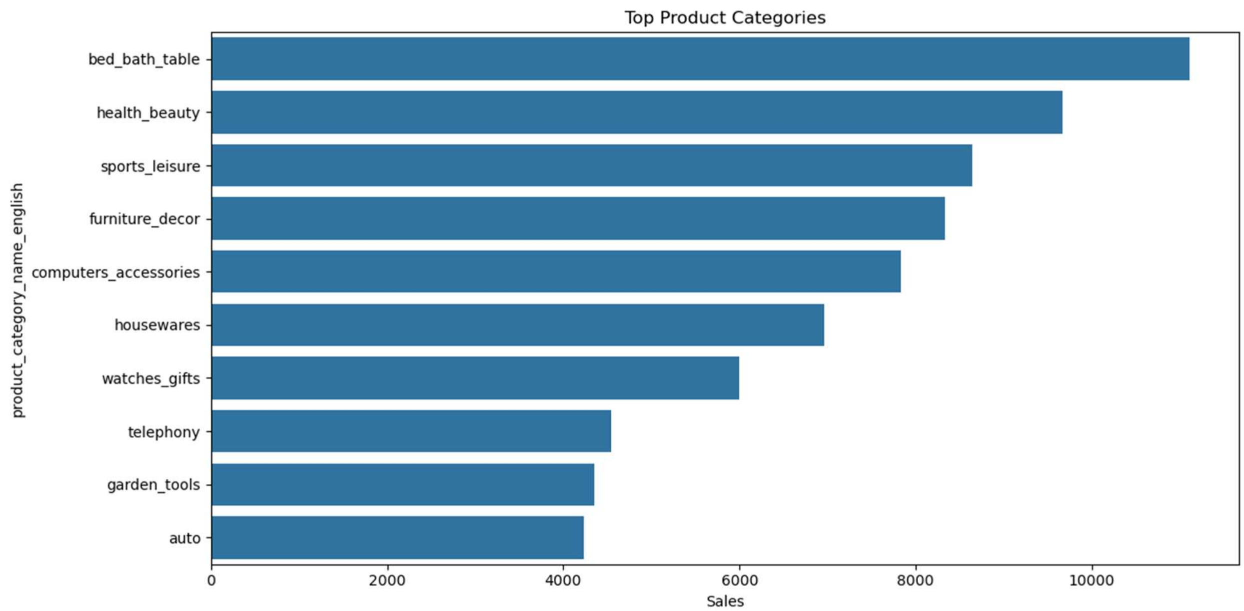
Visualization 1: Revenue by State



Description:

This chart illustrates revenue distribution across Brazilian states.

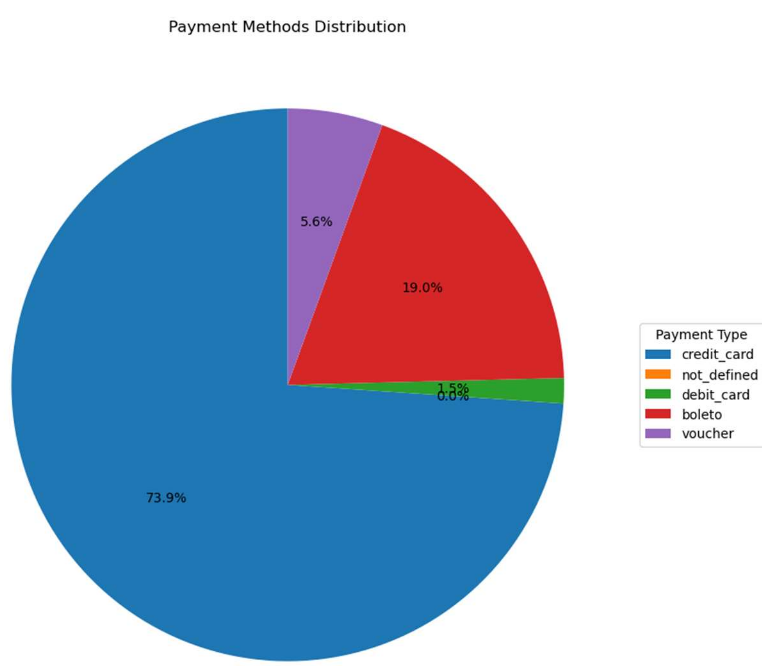
Visualization 2: Top Product Categories



Description:

This chart highlights the best-selling product categories.

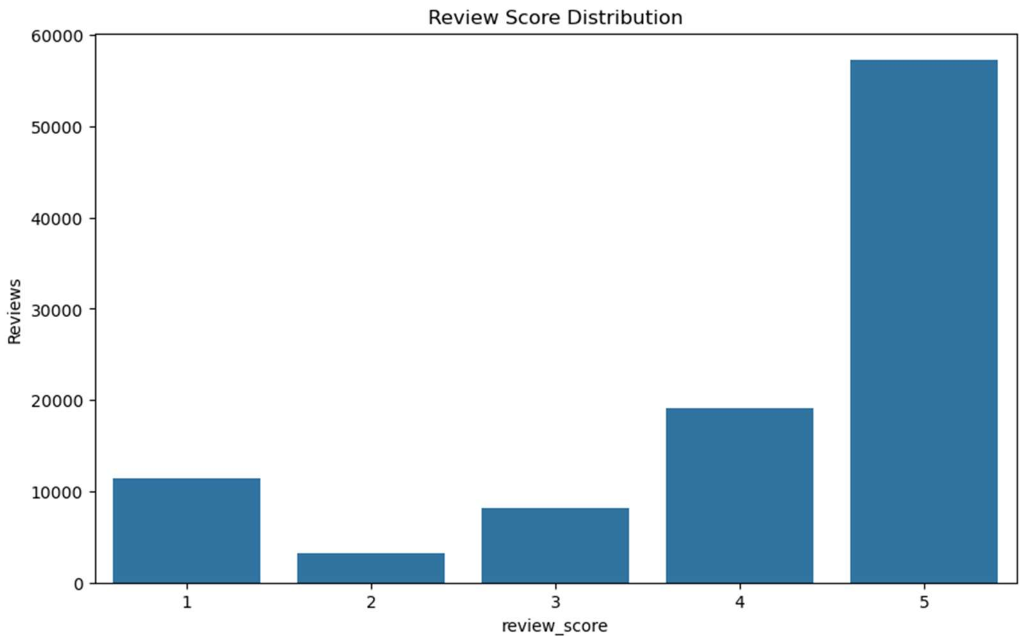
Visualization 3: Payment Method Distribution



Description:

This visualization shows customer payment preferences.

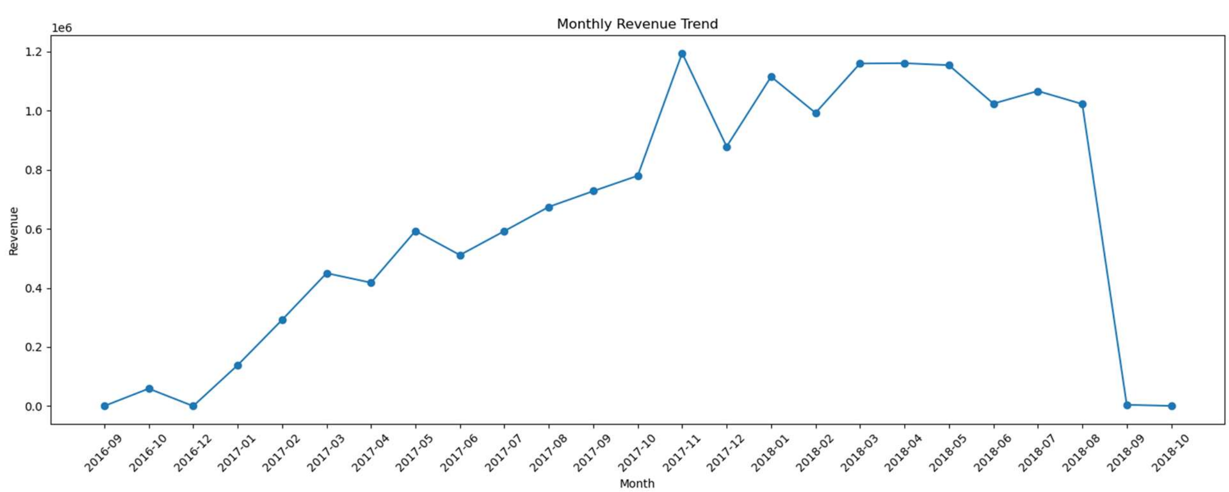
Visualization 4: Review Score Distribution



Description:

This chart displays customer review ratings.

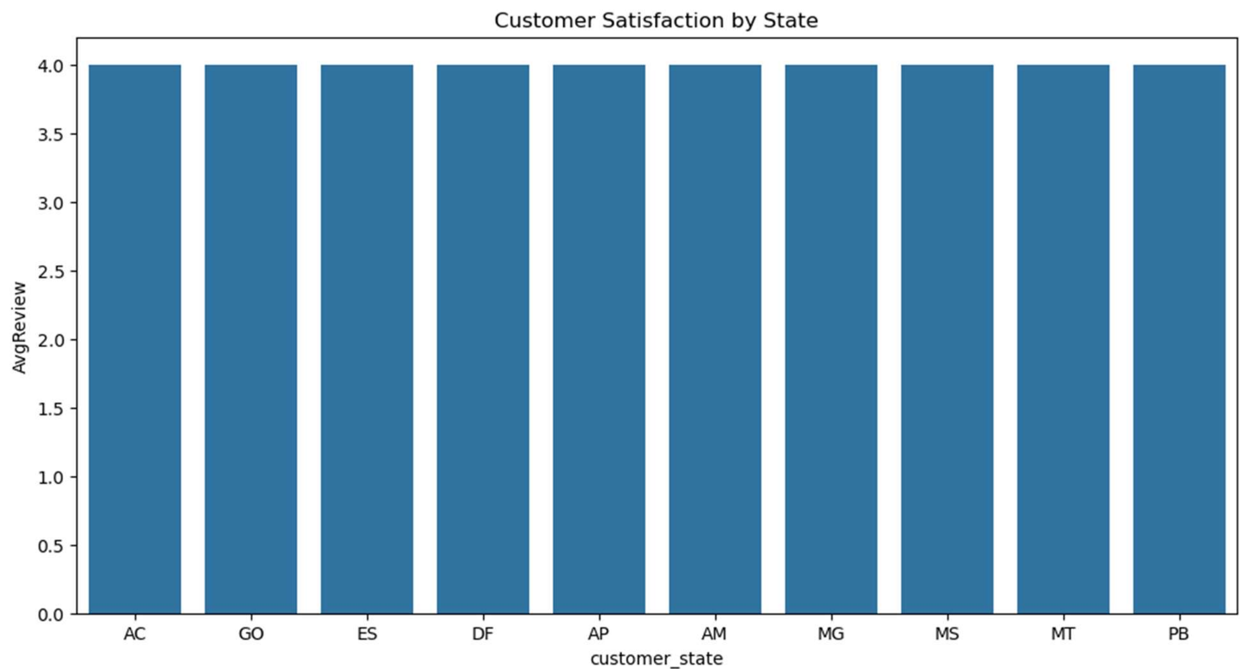
Visualization 5: Monthly Revenue Trend



Description:

This visualization presents revenue trends over time.

Visualization 6: Customer Satisfaction by State



Description:

This chart compares average customer satisfaction across states.

6. Findings

Based on the SQL analysis and visualizations, several important findings were identified:

1. Revenue is concentrated in a relatively small number of Brazilian states.
2. Certain product categories dominate total sales volume.
3. Credit cards are the most frequently used payment method.
4. Most customers provide positive review scores, indicating overall satisfaction.
5. Delivery performance has a direct impact on customer ratings.
6. Some sellers consistently outperform others in terms of revenue generation.
7. Monthly sales trends reveal periods of higher customer demand.

These findings can help businesses improve marketing strategies, logistics operations, and customer service quality.

7. Conclusion

This project successfully analyzed the Olist Brazilian E-Commerce dataset using Microsoft SQL Server and Python.

SQL JOIN and GROUP BY operations were used to extract valuable business insights from multiple related tables. Python visualizations helped communicate these insights clearly and effectively.

The analysis demonstrated how data-driven decision-making can improve understanding of customer behavior, product performance, seller efficiency, and operational effectiveness.

Future work could include machine learning models for sales prediction, customer segmentation, recommendation systems, and delivery performance forecasting.

8. References

Olist Brazilian E-Commerce Public Dataset : https://www.kaggle.com/datasets/olistbr/brazilian-ecommerce

Microsoft SQL Server Documentation : https://learn.microsoft.com/sql

Python Documentation : https://www.python.org

Pandas Documentation : https://pandas.pydata.org

Matplotlib Documentation : https://matplotlib.org

Seaborn Documentation : https://seaborn.pydata.org